Research Statement

[Ziyang Zhou]

Research Interests

My research interests center on advancing **Multi-Agent System** through **large language model** and their applications in scientific discovery and human-centric AI. I am particularly focused on developing **self-evolving agent systems** that can autonomously improve through experience, leveraging **agentic reinforcement learning** and **LLM post-training** techniques. My work spans **multi-agent collaboration frameworks**, **agent memory architectures**, and **retrieval-augmented generation (RAG)** systems that enable agents to reason over complex tasks in domains such as coding, deep research, and scientific problem-solving.

Beyond agent systems, I maintain strong interests in **audio and natural language processing**, particularly in emotion and empathy modeling, acoustic scene classification, and multimodal reasoning. I am driven by the vision of creating **AI systems that can learn, evolve, and collaborate** to tackle open-ended challenges across science, healthcare, and education. My goal is to contribute personalized AI and self-evovling agents to help realize **AGI** and evolve great agent products to solve human's issues.

Research Background and Contributions

My research has centered on developing **multi-agent systems** and **agent memory architectures** that enhance reasoning capabilities across diverse tasks. At **Yale University's Gerstein Lab** and **OPPO Personal AI**, I contributed to building the **Agent Knowledge Base (AKB)**, a cross-framework knowledge repository consolidating thousands of agent trajectories with human-curated and LLM-generated analyses. Through large-scale experiments across agent frameworks (smolagent, SWE-Agent, OpenHands, OWL) and benchmarks (GAIA, SWE-bench, GPQA), we demonstrated consistent performance gains of up to **+18.7pp in pass@3 on GAIA**, validating the effectiveness of cross-domain knowledge transfer for deep research tasks.

At **XJTLU's Human-Centered Language Computing Group**, I led the development of three successive **multi-agent frameworks for irony detection**: **CAF-I**, **SEVADE**, and **RAM-SD**. These frameworks progressively improved multi-agent reasoning through collaborative analysis, self-evolving evaluation mechanisms, and retrieval-augmented "Learn-Before-Reason" strategies, achieving up to **+7.01% Macro-F1 improvement** over GPT-4o baselines.

Currently, as an Algorithm Engineer Intern at **AxiomsTen**(from Jun. to Dec.), an AI hardware startup, I am leading the design and iteration of **agentic workflow and prompt systems** for personalized AI assistants. I have built FastAPI-based agent services for event-specific summarization and mindset extraction, and collaborated on designing a **Graph-RAG** abstraction layer for context reasoning and user event modeling.

Additionally, in collaboration with **Eigen AI**, I am developing **self-evolving agent frameworks** that autonomously generate and evaluate mcp tasks (such as online shopping), and designing data-centric systems that fine-tune open-source LLMs via **reinforcement learning (GRPO)** on task-specific data.

Beyond agent systems, I have contributed to **audio processing research** at XJTLU's Audio Intelligence Lab, designing the **AdapTF-SepNet framework** for acoustic scene classification with AudioSet-driven adaptive pre-training, achieving **+7.61% accuracy improvements** across device domains. I am proficient in **PyTorch, Linux, and ML/DL frameworks**, with extensive experience in developing end-to-end pipelines for complex multi-task learning scenarios.

Future Research Directions

In my future research, I plan to expand into the following areas:

- **Agentic Reinforcement Learning and Self-Evolving Systems**: I aim to develop agent frameworks that autonomously improve through self-data generation and Agent Reinforcement Learning. Building on my work with self-evolving coding agents and mcp tasks agents, I want to explore how agents can iteratively refine their reasoning strategies, tool usage, and knowledge bases through continuous interaction with complex environments. This includes designing reward mechanisms that capture multi-dimensional success criteria beyond task completion, such as code quality, reasoning coherence, and environment interactions.

- **LLM Post-Training for Agent Capabilities**: I am interested in advancing **post-training techniques** that specialize LLMs for agentic tasks. My experience with GRPO-based fine-tuning on task-specific data has demonstrated the potential of data-centric approaches. I plan to investigate how synthetic data generation, preference learning, and multi-task curriculum design can enhance agent capabilities in planning, tool use, memory retrieval, and long-horizon reasoning. This includes exploring efficient fine-tuning methods that balance general intelligence with domain-specific expertise.

- **Agent Memory and Long-Term Reasoning**: Building on my work with retrieval-augmented agent memory and Graph-RAG systems, I aim to develop **adaptive memory architectures** that support long-term planning and contextual reasoning. This includes investigating how agents can selectively consolidate, retrieve, and update knowledge across sessions, and how memory mechanisms can facilitate transfer learning across diverse task domains. I am interested in combining episodic memory (specific experiences), semantic memory (general knowledge), and procedural memory (skills and strategies) to create more robust and generalizable agent systems.

- **AI for Science, Healthcare, and Human-Centric Applications**: I plan to apply agent systems to domains where autonomous reasoning and adaptive behavior can create significant impact, such as scientific discovery, medical diagnosis support, personalized emotion replies and personalized education. This includes developing agents that can assist researchers in literature exploration, hypothesis generation, and experimental planning, as well as healthcare assistants that provide context-aware, empathetic interactions. My work on emotion and empathy modeling provides a foundation for building human-centric AI systems that balance task performance with user experience and trust.

I am eager to contribute to these research directions through rigorous experimentation, interdisciplinary collaboration, and the development of scalable, generalizable solutions that advance the frontier of autonomous intelligent systems.

Personal Strengths

I believe that my greatest strength in research lies in the combination of technical depth, research insights, and rapid prototyping ability. My experience developing multiple successive agent frameworks has taught me to identify limitations in existing approaches and iteratively design solutions that address fundamental challenges. I excel at bridging theoretical concepts with practical implementation, as demonstrated by my work integrating retrieval-augmented memory, self-evolving mechanisms, and reinforcement learning into functional agent systems.

One of my key strengths lies in my ability to conduct large-scale empirical research across diverse benchmarks and frameworks. My work validating the Agent Knowledge Base across smolagent, SWE-Agent, OpenHands, and OWL under GAIA, SWE-bench, and GPQA demonstrates my capability to design comprehensive experiments that yield generalizable insights. I am proficient in the full research pipeline, encompassing data curation, model training, evaluation, and analysis, and have hands-on experience with widely used LLM inference benchmarks.

Furthermore, I maintain a forward-thinking perspective and keen insight into emerging technologies in the rapidly evolving field of LLM agents. My industrial experience at AxiomsTen has strengthened my ability to translate research insights into production-grade systems, while my academic collaborations have honed my skills in identifying and tackling fundamental research questions. In team projects, I excel at synthesizing complex information, identifying core challenges, and proposing constructive solutions that balance theoretical rigor with practical feasibility. What's more, I keep a close connection with the latest agent research team around the world, which can provide me continuous research insights and direction for the development.

My international educational and collaborative experiences provide me with a diverse perspective and allow me to work efficiently in global research environments. This openness to ideas and my passion for continuous learning enable me to contribute unique insights and drive meaningful progress in the field of autonomous intelligent systems. In my free time, I am also kind to help with those students who are curious with my work and need some guidance for their research direction.